

MATMEK-4270 Assignment 1

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1 Exact solution to the 2D wave equation

The wave equation is given as

$$\frac{\partial^2 u}{\partial t^2} = c^2 \nabla^2 u. \quad (1)$$

To show that the solution

$$u(t, x, y) = e^{i(k_x x + k_y y - \omega t)} \quad (2)$$

satisfies this equation, we first find partial derivatives with respect to all variables:

$$\begin{aligned} \frac{\partial u}{\partial t} &= -i\omega u \\ \frac{\partial^2 u}{\partial t^2} &= i^2 \omega^2 u = -\omega^2 u \\ \frac{\partial u}{\partial x} &= i k_x u \\ \frac{\partial^2 u}{\partial x^2} &= i^2 k_x^2 u = -k_x^2 u \\ \frac{\partial u}{\partial y} &= i k_y u \\ \frac{\partial^2 u}{\partial y^2} &= i^2 k_y^2 u = -k_y^2 u \end{aligned}$$

and substitute them into eq. 1:

$$\begin{aligned} \frac{\partial^2 u}{\partial t^2} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) &\Rightarrow -\omega^2 u = c^2 (-k_x^2 u - k_y^2 u) \Rightarrow \\ \omega^2 &= c^2 (k_x^2 + k_y^2). \end{aligned} \quad (3)$$

We arrive at a conclusion that solution 2 satisfies the wave equation as long as the relation 3 between parameters holds.

2 Dispersion coefficient

Given discretized equation:

$$\frac{u_{i,j}^{n+1} - 2u_{i,j}^n + u_{i,j}^{n-1}}{\Delta t^2} = c^2 \left(\frac{u_{i+1,j}^n - 2u_{i,j}^n + u_{i-1,j}^n}{h^2} + \frac{u_{i,j+1}^n - 2u_{i,j}^n + u_{i,j-1}^n}{h^2} \right)$$

we rearrange it to get:

$$u_{i,j}^{n+1} - 2u_{i,j}^n + u_{i,j}^{n-1} = \frac{c^2 \Delta t^2}{h^2} (u_{i+1,j}^n - 2u_{i,j}^n + u_{i-1,j}^n + u_{i,j+1}^n - 2u_{i,j}^n + u_{i,j-1}^n)$$

and substitute the assumed CFL number $C = 1/\sqrt{2}$ producing:

$$2(u_{i,j}^{n+1} - 2u_{i,j}^n + u_{i,j}^{n-1}) = u_{i+1,j}^n - 2u_{i,j}^n + u_{i-1,j}^n + u_{i,j+1}^n - 2u_{i,j}^n + u_{i,j-1}^n.$$

We then subtract $-4u_{i,j}^n$ from both sides:

$$2(u_{i,j}^{n+1} + u_{i,j}^{n-1}) = u_{i+1,j}^n + u_{i-1,j}^n + u_{i,j+1}^n + u_{i,j-1}^n$$

and proceed by replacing u with the discretized exact solution:

$$\begin{aligned} \text{LHS} &= 2 \left(e^{i(kh(i+j) - \tilde{\omega}(n+1)\Delta t)} + e^{i(kh(i+j) - \tilde{\omega}(n-1)\Delta t)} \right) = 2e^{ikh(i+j)} \left(e^{-i\tilde{\omega}(n+1)\Delta t} + e^{-i\tilde{\omega}(n-1)\Delta t} \right) = \\ &= 2e^{ikh(i+j)} e^{-i\tilde{\omega}n\Delta t} (e^{-i\tilde{\omega}\Delta t} + e^{i\tilde{\omega}\Delta t}) \end{aligned}$$

$$\begin{aligned} \text{RHS} &= e^{i(kh(i+1+j) - \tilde{\omega}n\Delta t)} + e^{i(kh(i-1+j) - \tilde{\omega}n\Delta t)} + e^{i(kh(i+j+1) - \tilde{\omega}n\Delta t)} + e^{i(kh(i+j-1) - \tilde{\omega}n\Delta t)} = \\ &= 2 \left(e^{i(kh(i+j+1) - \tilde{\omega}n\Delta t)} + e^{i(kh(i+j-1) - \tilde{\omega}n\Delta t)} \right) = 2e^{-i\tilde{\omega}n\Delta t} \left(e^{ikh(i+j+1)} + e^{ikh(i+j-1)} \right) = \\ &= 2e^{-i\tilde{\omega}n\Delta t} e^{ikh(i+j)} (e^{ikh} + e^{-ikh}) \end{aligned}$$

$$\begin{aligned} \text{LHS} &= \text{RHS} \\ 2e^{ikh(i+j)} e^{-i\tilde{\omega}n\Delta t} (e^{-i\tilde{\omega}\Delta t} + e^{i\tilde{\omega}\Delta t}) &= 2e^{-i\tilde{\omega}n\Delta t} e^{ikh(i+j)} (e^{ikh} + e^{-ikh}) \\ e^{-i\tilde{\omega}\Delta t} + e^{i\tilde{\omega}\Delta t} &= e^{-ikh} + e^{ikh} \end{aligned}$$

which implies that:

$$\tilde{\omega}\Delta t = kh \quad \Rightarrow \quad k = \frac{\tilde{\omega}\Delta t}{h}. \quad (4)$$

Now, using the assumption that $k_x = k_y = k$, we substitute relation 4 to equation 3 producing:

$$\omega^2 = c^2 (k_x^2 + k_y^2) \quad \Rightarrow \quad \omega^2 = 2c^2 k^2 \quad \Rightarrow \quad \omega = \sqrt{2} \frac{c\Delta t}{h} \tilde{\omega}.$$

Finally, we once again use the fact that $C = \frac{1}{\sqrt{2}} = \frac{c\Delta t}{h}$, which brings us to the conclusion:

$$\omega = \tilde{\omega}.$$